

Microscopy of Surface Microstructural Phenomena in 8620 Steel After Annealing

Kaleb Hood, Sebastian Lara, Samuel Olson, Martin Silva, Jun Jiao



Meeting-report

Microscopy of Surface Microstructural Phenomena in 8620 Steel After Annealing

Kaleb Hood¹, Sebastian Lara¹, Samuel Olson¹, Martin Silva¹, and Jun Jiao^{1,*}

¹Department of Mechanical and Materials Engineering, Portland State University, Portland, OR, USA

*Corresponding author: jjiao@pdx.edu

This report focuses on the effects of annealing on the surface microstructures of a high-iron content steel for the support of the direct growth of graphene. Initially, 8620 steel is composed of ferrite and pearlite grains, and a polished surface appears rough. After annealing in an H₂ and Ar environment, the surface became smoother, grain size increased, and grain morphology changed. In contrast, annealing under H₂ and C₂H₂ resulted in unchanged grain size and graphene precipitation on the polished smooth surface. Secondary electron (SE) and backscattered electron (BSE) images allowed for identification of ferrite and pearlite in the original microstructure [1]. SEM was used to compare the surface structures after the two annealing environments, H₂ + Ar or H₂ + C₂H₂. The sample annealed in H₂ and C₂H₂ had graphene form on the surface. Raman spectroscopy was used to characterize the multilayer graphene (MLG) coating, with few-layer graphene (FLG) and single layer graphene (SLG) in coating gaps. These methods demonstrate the annealing environment's influence on 8620 steel surfaces, to improve the growth of graphene on these surfaces toward the achievement of high-quality coatings.

Samples of 8620 steel were polished with 120, 400, 800, and 1200 grit sandpapers, followed by 3 μ m polycrystalline diamond. SEM micrographs were acquired with an SNE 4500M Plus. Raman spectra were acquired with a Horiba Jobin Yvon HR800. Annealing was at 650°C for 60 minutes, under H₂ + Ar or H₂ + C₂H₂; H₂ is used for its active role in annealing and graphene growth processes [2].

The polished 8620 steel surfaces with original or annealed surface grain structures are shown in Fig. 1. In Fig. 1(a), the SE image shows the polished, rough, surface and the BSE image shows the original ferrite and pearlite grain structure (grain size 10-15 μ m), with pearlite having bright contrast due to high-density cementite lamellar structure (see inset SE image of typical pearlite). Fig. 1(b) shows the smoothed surface after annealing for 60 minutes under Ar and H₂. There are both large single grains (25-50 μ m) or collections of smaller grains (5-10 μ m). It is likely that during annealing pearlite decomposes, grains recrystallize, and the composition becomes more uniform, as shown by Fig. 1(b) in the BSE image [3]. Excess carbon precipitates as carbides on the surface as shown by the inset of a high magnification SE image in Fig. 1 (b). By adding carbon via C₂H₂, the precipitation instead forms graphene as shown in Fig. 1(c), with a grain size around 5-15 μ m. The zoomed-in image in Fig. 1(d), shows the largely MLG coating, with gaps between covered in FLG and SLG. Some carbides formed at grain boundaries. Representative Raman spectra of MLG and SLG regions are shown in Fig. 1 (e). The I_{D:G} ratio represents disorder in graphene, and the I_{2D:G} ratio represents graphene thickness where I_{2D:G} > 1 is SLG. The MLG spectrum in green, has a low I_{D:G} < 0.05, indicating high quality, and an I_{2D:G} of 0.49. An example of a SLG spectrum in blue has a higher I_{D:G} of 0.4 indicating some disorder, however the I_{2D:G} of 1.6 is a clear sign of SLG.

Raman map arrays (10x10 μ m grid, 0.5 μ m step size) in Fig. 2 show the MLG coating with FLG and SLG in the gaps. An optical image in Fig. 2(a) shows the green MLG islands, and light-colored gaps. Fig. 2(b) is the I_{2D:G} map, with MLG shown as blue pixels, and FLG (teal) or SLG (green) covering the gap. The I_{D:G} map is shown in Fig. 2(c), where green represents a very low I_{D:G} of < 0.1. The thin graphene has a slightly higher I_{D:G}. A histogram of I_{2D:G} from the mapped area in Fig. 2(d) shows that most of the coating in the map is MLG, and the histogram of I_{D:G} in Fig. 2(e) shows the coating mostly has I_{D:G} < 0.1.

By annealing polished 8620 steel at 650°C under Ar and H₂, the surface is smoothed and the grain structure changes from one of pearlite and ferrite to a more uniform structure with larger grains. When annealing under C₂H₂ and H₂, the surface is covered with a graphene coating and the grain sizes are closer to the pre-annealed size. The use of H₂ and C₂H₂ or other hydrocarbon gases during annealing shows promise for the development of graphene coatings on the surfaces of high iron content steels.

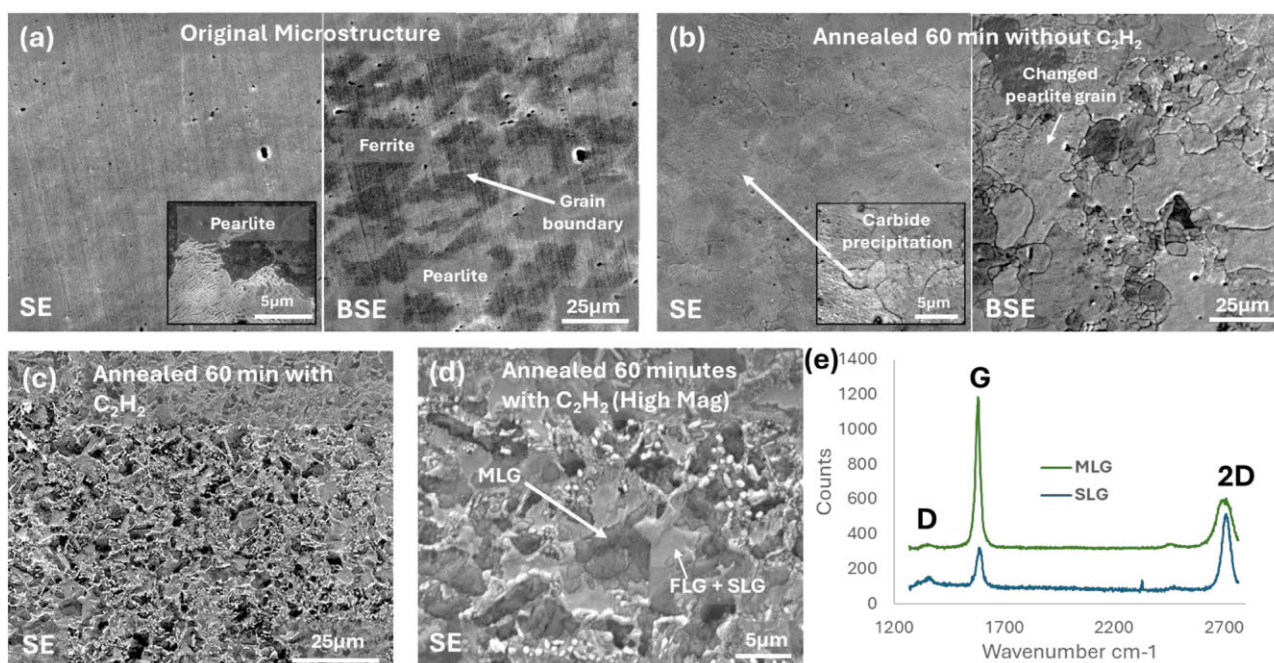


Fig. 1. SEM analysis of 8620 steel under annealed/unannealed conditions; (a) SE image of a polished surface with inset of a typical pearlite structure. The BSE image shows pearlite and ferrite; (b) SE image of 8620 surface after 60 min annealing without C_2H_2 , inset shows textured carbides on surface and BSE shows a more uniform composition than before annealing; (c) SE image of graphene formation on 8620 annealed for 60 min with C_2H_2 ; (d) high mag image of (c) where MLG, FLG and SLG are visible with some carbide precipitates; (e) Representative Raman spectra of the MLG and SLG from (d)

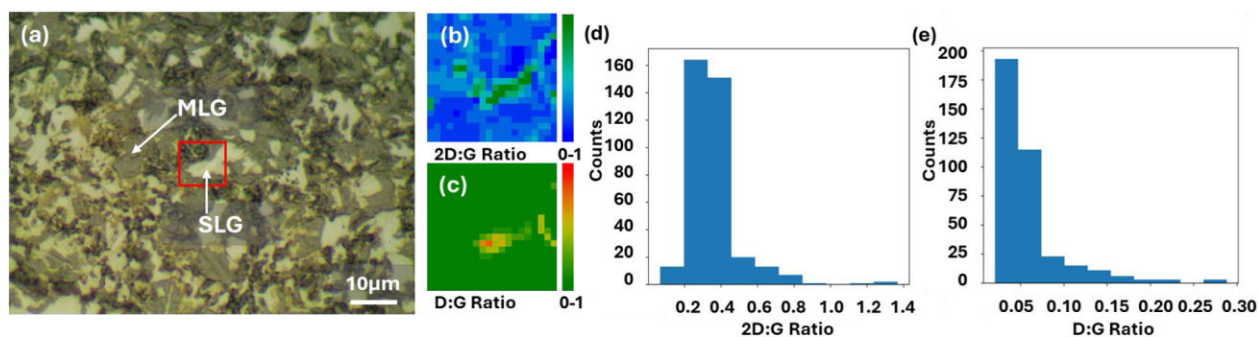


Fig. 2. Raman maps of graphene coating; (a) optical image of the large area covered by graphene. The red square indicates the mapped areas shown in (b) and (c); (b) $I_{2D:G}$ map showing MLG coating and FLG or SLG in gaps; (c) $I_{D:G}$ map showing overall low disorder in the coating; (d) histogram of $I_{2D:G}$ for mapped area; (e) histogram of $I_{D:G}$ for mapped area.

References

1. J Steele, *Microscopy and Microanalysis* 9(S2) (2003). <https://doi.org/10.1017/s1431927603440610>
2. Lampert *et al.*, *Journal of Physical Chemistry C* 120(46) (2016), p. 26498. <https://doi.org/10.1021/acs.jpcc.6b06459>
3. Lara-Guevara *et al.*, *Journal of Materials Research* 32(11) (2017), p. 2202. <https://doi.org/10.1557/jmr.2017.47>